



SUPERCARS – 2017 TIRE TEST



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Revision	Revision Date	Description
1	27JAN2017	Initial Release

1. Background

This document presents the details of a tire testing project carried out for Supercars at the Tire Testing Laboratory at Calspan in Buffalo, NY, USA (Calspan Tire Research Project 1792). First an overview of the test facility and the measurements are given. Then, an overview of the structure of the collected data and other deliverables are provided. Finally, details around tire test procedures and test conditions are presented.

This document has not been written to determine the results or make any kind of data analysis. It has been written to provide an introduction to the tire testing project, the different measurements and test methods, and how to navigate the data set.

Calspan would like to thank Mark Bryant, Justin Lin and Charlie Ping, for the technical input and support they provided for this project.

2. Test Equipment and Setup

2.1 Test Equipment

Testing was carried out on Calspan's tire testing machine, which is a continuous flat belt machine. The machine is capable of testing tires at speeds in excess of 320 km/h and at loads up to 50 kN. The machine is very well suited to evaluate the performance of motorsports tires.

During testing, the test tire is mounted to an articulated head that controls the position of the tire with respect to the roadway. By varying the tire orientation different vertical loads, slip angles and inclination angles (camber angles) can be achieved. Driving and braking torques can also be fed from an electric drive motor directly to the wheel/tire. The roadway consists of a 10 meter long continuous steel belt that is covered with a 120-grit *Three-M-ite* material. The loads of the tire are being reacted by a high-pressure air bearing under the contact patch. The tire inflation pressure is continuously measured and controlled during testing through air hoses.



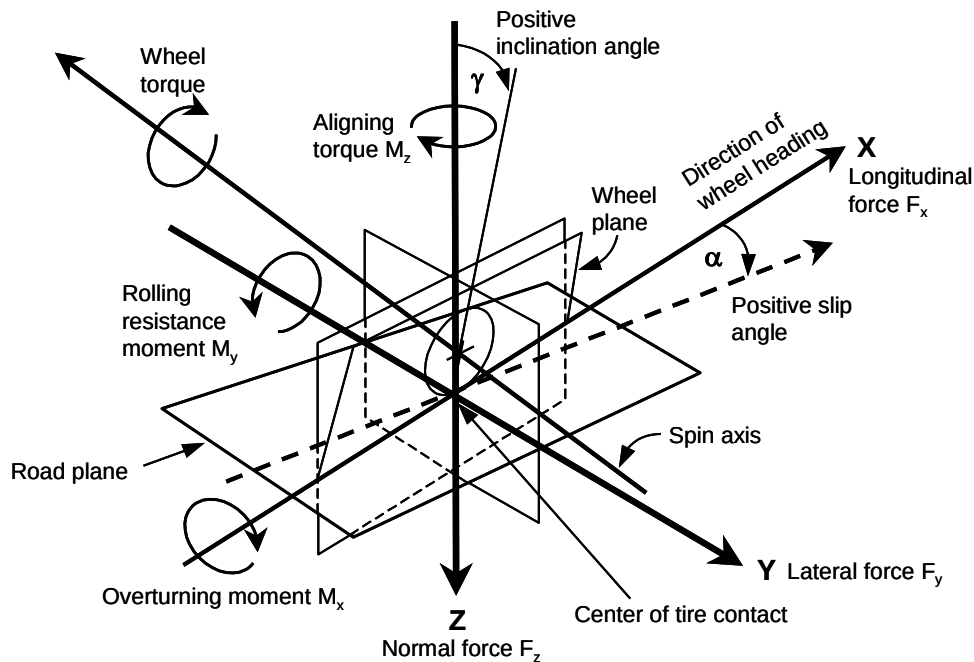
2.2 Test Execution

The testing is divided into runs. For each run, a tire is fitted to the machine and is put through a series of conditions. This series of conditions is called the test procedure. The test procedures are designed to collect different types of tire performance data.

2.3 Tire Data

All tire data in this project was collected in and are delivered in the SAE tire axis system according to the SAE J2047 standard. The right-handed coordinate system used is thus:

- X-axis is positive in the forward direction
- Y-axis is positive to the right
- Z-axis is positive downwards



For a tire mounted on the right hand side of a vehicle, the following angular relationships apply:

- When the vehicle is turning right the tire slip angle is negative
- When the vehicle is turning left the tire slip angle is positive
- Negative inclination angle is equal to negative camber, meaning the top of the tire is leaning towards the center of the vehicle.

2.4 Measurements

During testing a range of different channels are recorded. Data signals were recorded at 100 Hz after going through a 10 Hz low-pass filter. The 38 different data channels are presented in the table below. Note that not all channels are available for all test runs depending on the machine configuration and test type. Additional notes for some of the channels can be found below the table.

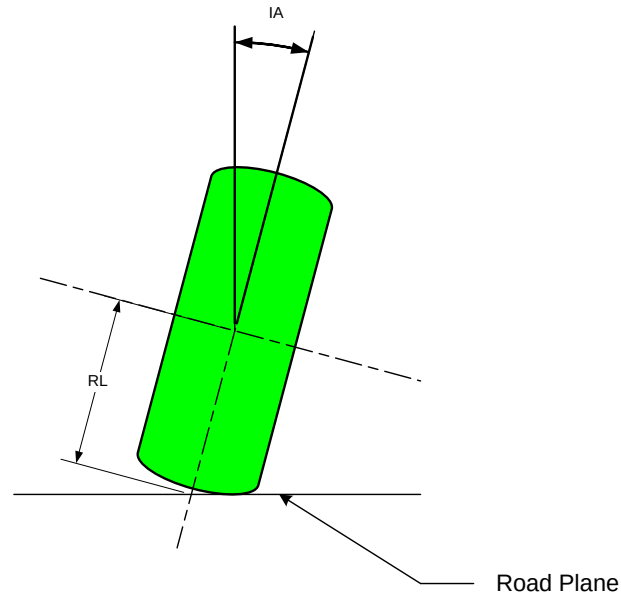
2.4.1 Measurement Channels

Acronym	Unit	Description
ET	s	Elapsed Time from start of test
V	kph	Roadway velocity
SA	deg	Slip angle
IA	deg	Inclination angle (camber angle)
RL	cm	Loaded Radius (see note #1)
FX	N	Longitudinal Force
FY	N	Lateral Force
FZ	N	Vertical Force
MX	Nm	Overturning Moment
MZ	Nm	Self-Aligning Moment
NFX	none	Normalized Longitudinal Force
NFY	none	Normalized Lateral Force
P	psi	Inflation Pressure
N	rpm	Wheel Rotational Velocity
HT	Nm	Wheel Torque
RE	cm	Effective Rolling Radius
SR	SAE	Slip Ratio (SAE) (see note #2)
SL	-	Slip Ratio (see note #2)
TSTO	deg c	Tire Surface Temperature (Outboard Tread) (note #3)
TSTC	deg c	Tire Surface Temperature (Center Tread)
TSTI	deg c	Tire Surface Temperature (Inboard Tread)
TSTSI	deg c	Tire Surface Temperature (Inboard Sidewall)
RST	deg c	Road Surface Temperature
CFZ	N	Commanded Vertical Load
CV	kph	Commanded Roadway Velocity
CSA	deg	Commanded Slip Angle
CSR	none	Commanded Slip Ratio
CP	bar	Commanded Inflation Pressure
CIA	deg	Commanded Inclination Angle
AmbTmp	deg c	Ambient Temperature
TLT1	deg c	Tire Inner Liner Temperature (most inboard) (see note #4)
TLT2	deg c	Tire Inner Liner Temperature
TLT3	deg c	Tire Inner Liner Temperature
TLT4	deg c	Tire Inner Liner Temperature
TLT5	deg c	Tire Inner Liner Temperature
TLT6	deg c	Tire Inner Liner Temperature
TLT7	deg c	Tire Inner Liner Temperature
TLT8	deg c	Tire Inner Liner Temperature (most outboard)
BELTX	cm	Belt Travel Distance (used in longitudinal stiffness test)

2.4.2 Measurement Notes

Note 1: Loaded Radius

During testing, loaded radius is measured, however, often vehicle engineers are interested in the axle height (vertical distance from wheel center to road plane). The relationship between loaded radius and axle height can be seen in the illustration below. The difference between the two values become larger with larger camber angles.



Note 2: Slip Ratio

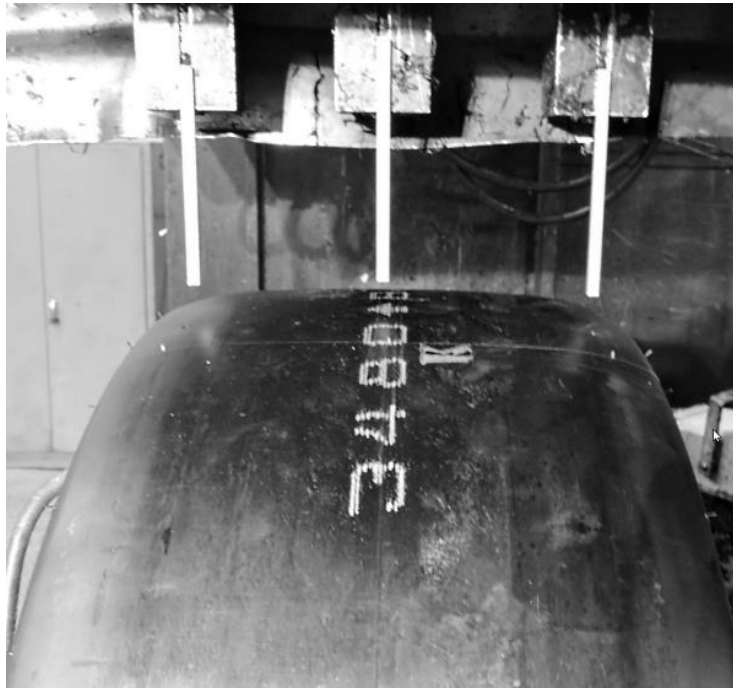
The Brake/Drive data contains two different slip ratio channels, SR and SL. SR is a definition of slip ratio that uses the loaded radius. Since loaded radius can be measured in real-time during testing, this definition is used in the real-time control of the tire testing machine. When SR is zero, the longitudinal force may not be zero. After testing is completed, the data is post-processed to calculate the other definition, SL. This definition uses the effective rolling radius to calculate slip ratio. The effective rolling radius is determined in a free-rolling state ($F_X=0$) and as such when SL is zero the longitudinal force is zero (or close to).

For analysis of Brake/Drive data it is strongly recommended to use the SL channel.

For further discussion on the different definitions of slip ratio, please see section 14.2 of the book Race Car Vehicle Dynamics in which differences between SR and SL are presented.

Note 3: External Tire Temperatures

Tire tread temperatures were measured at three locations across the tire tread. The locations of each measurement are shown in the photo below. The metal bars show the alignment of each sensor, with the inboard sensor being to the right in the photo. The external sidewall temperature is also measured on the inboard sidewall (not shown in photo).



Note 4: Internal Tire Temperatures

For some of the test runs, an internal IR sensor was used to measure the temperature of the tire inner liner. Eight measurement points are evenly distributed along an approximately 150 mm long line, centered on the tire's centerline. Channel 1 (TLT1) is furthest towards the inboard shoulder, Channel 8 (TLT8) is furthest towards the outboard shoulder. Channel 4/5 (TLT4/5) are approximately aligned with the external temperature channel TSTC. Note that the internal temperature measurements have a higher noise level compared to the external temperature measurements. This is due to a different measurement technology being used for the internal tire temperatures.

For runs where the internal temperatures were not measured, the value of all TLT channels are reported as zero in the data files.

3. Data Overview

3.1 Data Deliverables

For this testing project, Calspan has delivered a number of different items. Details of the deliverables can be found below.

3.1.1 Measurement Data

All raw data is delivered in tab-separated ASCII format (*.dat). This data can be opened in a range of different analysis software, such as Excel, MATLAB and Python. It is also supported by most commercial tire analysis and modelling software.



Two types of files are supplied, “raw” and “run” files. They can be differentiated by their names being either A_1792rawXX.dat or A_1792runXX.dat.

Raw: The “raw” file contains time-continuous data for the entire run. No additional post-processing is done to this file. This file is typically used to investigate overall structure of a test, the warmup or thermal trends.

Run: The “run” file has been post-processed (for example to calculate SL and remove transient sweep effects from the slip angle sweeps). Additionally it only presents data from relevant test conditions (such as a slip angle sweep). As a result, the time channel will have gaps in it. This file is typically used for tire force & moment performance analysis, plotting and modelling.

Several of the Free-Rolling tests, had additional measurement sections at the end of the test procedure (for example specific Vertical Stiffness measurements). Those measurement sections have been separate to their own run files, using a suffix such as “_VertStiff”.

3.1.2 Summary Table

The *Summary Tables* Excel document presents details of the test run order as well as additional measurements. The document contains the following sheets:

1. Cornering Test Schedule
2. Brake/Drive Test Schedule
3. Tire Identification Sheet
4. Tire Code Summary
 - a. Presents test runs organized by tire type
5. Spring Rates, Cornering Stiffness
 - a. Automatically calculated tire metrics from slip angle sweeps
6. Tire Weights
 - a. Measured on un-mounted tire before and after test runs
7. Tread Depths
 - a. Measurements at wear marker. Note that due to pick up the measurements in some cases show increased tread depths after the run
8. Tread Temperatures
 - a. Pyrometer measurements taken typically within 60 seconds of test completion

3.1.3 Post-Test Photos

These photos are taken at the end of each test run and may highlight differences in tire wear and compound behavior. Note that the type of test procedure may influence the visual appearance of the tire.

3.1.4 Test Video

Video files showing the test runs. The files can be opened in most media players. For some test runs a strobe light was used to freeze the tire to provide a more detailed view of the tire tread surface. Note that video files that used strobe lights do not play back in real time.



3.1.5 Infrared Camera Video

An infrared camera (model: FLIR A325c) was used to capture detailed temperature data. The data files (.seq) can be opened using the free software "FLIR Tools" which can be downloaded here: <http://www.flir.com/instruments/display/?id=54865>

3.2 Tire Test Run Order

The Tire Test Run Order is presented on the next page. For more complete information please refer to the 1792 Project Summary document.

In the run order, note that three different tire types were tested, allowing for comparisons of compounds as well as comparison of old and new tire constructions.

Manufacturer	Size	Description	Code	Type
Dunlop	290/675 R18	Soft-XP	3480J	2017 Soft
Dunlop	290/675 R18	Hard-XP	3480K	2017 Hard
Dunlop	280/680 R18	SPORT MAXX	3024	2016 Hard

Run	Test Type	Tire	Procedure Name	TLT	Note
1792-1	FreeRoll	2017 Soft	SC_FreeRoll_27psi Warmup	x	Check Test of Warm-Up
1792-2	FreeRoll	2017 Soft	SC_FreeRoll_24psi_wThermal	x	
1792-3	FreeRoll	2017 Soft	SC_FreeRoll_24psi_wThermal	x	
1792-4	FreeRoll	2017 Soft	SC_FreeRoll_27psi_wIA	x	
1792-5	FreeRoll	2017 Soft	SC_FreeRoll_30psi_wVertStiff		
1792-6	FreeRoll	2017 Soft	SC_FreeRoll_33psi_wExtraIA		
1792-7	FreeRoll	2017 Soft	SC_FreeRoll_wDifferentSpeeds	x	
1792-8	FreeRoll	2017 Soft	SC_FreeRoll_21psi		
1792-9	FreeRoll	2017 Hard	SC_FreeRoll_27psi_wIA	x	
1792-10	FreeRoll	2017 Hard	SC_FreeRoll_24psi_wThermal		
1792-11	FreeRoll	2017 Hard	SC_FreeRoll_30psi_wVertStiff		
1792-12	FreeRoll	2016 Hard	SC_FreeRoll_27psi_wIA	x	
1792-13	FreeRoll	2016 Hard	SC_FreeRoll_24psi_wThermal		
1792-14	FreeRoll	2016 Hard	SC_FreeRoll_30psi_wVertStiff		
1792-15	BrkDrv	2017 Soft	SC_BrakeDrive_27psi Warmup	x	Check Test of Warm-Up
1792-16	BrkDrv	2017 Soft	SC_BrakeDrive_27psi	x	
1792-17	BrkDrv	2017 Soft	SC_BrakeDrive_24psi		
1792-18	BrkDrv	2017 Soft	SC_BrakeDrive_30psi		
1792-19	BrkDrv	2017 Soft	SC_BrakeDrive_33psi		
1792-20	BrkDrv	2017 Soft	SC_BrakeDrive_21psi		
1792-21	BrkDrv	2017 Hard	SC_BrakeDrive_27psi	x	
1792-22	BrkDrv	2017 Hard	SC_BrakeDrive_24psi		
1792-23	BrkDrv	2017 Hard	SC_BrakeDrive_30psi		
1792-24	BrkDrv	2016 Hard	SC_BrakeDrive_27psi	x	
1792-25	BrkDrv	2016 Hard	SC_BrakeDrive_24psi		
1792-26	BrkDrv	2016 Hard	SC_BrakeDrive_30psi		
1792-27	BrkDrv	2017 Soft	SC_RelaxSRsine		
1792-28	BrkDrv	2017 Soft	SC_StaticLong		
1792-29	BrkDrv	2017 Soft	SC_StaticLong_LowPressure		
1792-30	BrkDrv	2017 Soft	SC_BrakeDrive_wDifferentSpeed		
1792-31	BrkDrv	2017 Soft	SC_TheGrate		Check Test of procedure
1792-32	BrkDrv	2017 Soft	SC_TheGrate_WUP		
1792-33	BrkDrv	2017 Soft	SC_TheGrate	x	
1792-34	BrkDrv	2017 Soft	SC_TheGrate_WUP		
1792-35	BrkDrv	2017 Soft	SC_TheGrate		
1792-36	BrkDrv	2017 Hard	SC_TheGrate_WUP		
1792-37	BrkDrv	2017 Hard	SC_TheGrate		
1792-38	BrkDrv	2016 Hard	SC_TheGrate_WUP		
1792-39	BrkDrv	2016 Hard	SC_TheGrate		
1792-40	FreeRoll	2017 Soft	SC_ThermalOnly_24psi	x	Thermal test on cold tire
1792-41	FreeRoll	2017 Soft	SC_FreeRoll_27psi_wThermal		Testing on heat-cycled tire
1792-42	FreeRoll	2017 Hard	SC_FreeRoll_27psi_wThermal		Testing on heat-cycled tire
1792-43	FreeRoll	2016 Hard	SC_FreeRoll_27psi_wThermal		Testing on heat-cycled tire

4. Test Procedure Overview

A number of different types of test procedures were used to evaluate different aspects of tire performance. A few general notes about the test procedures:

- The warm-up procedures used for Free-Rolling and Brake/Drive tests are all done at the same test condition. This allows for analysis of tire repeatability.
- Throughout the test procedures “check sweeps” are performed. These are repetitions of a given test condition at different times in the test. The check sweeps can be used to evaluate how tire performance is changing due to factors such as temperature and wear.

4.1 Thermal Logic

To ensure consistent tire temperatures throughout a test procedure, the tire is moved into a pause condition after each sweep to allow high temperatures to dissipate. The tire is kept in this condition until a target temperature of 75°C is measured on the TSTC channel. Due to different thermal responses in different runs, the duration of the thermal pauses vary. As such, the overall length of different test runs vary.

4.2 Free-Rolling Test Procedures

The main free-rolling procedures consist of:

1. Warm-Up Section (at 27 psi)
2. Force & Moment Section (at different pressures)
3. Additional Measurement Section (one of the following)
 - a. Thermal: Tests tire at different temperature and sliding velocity conditions
 - b. IASweep: Tests detailed camber response at different conditions
 - c. VertStiff: Tests loaded radius and effective loaded radius behavior in a range of different conditions
 - d. ExtraIA: Extended camber response tests with higher slip angle and higher max camber angle

4.3 Brake/Drive Test Procedures

The main Brake/Drive procedures consist of:

1. Warm-Up Section (at 27 psi)
2. Force & Moment Section (at different pressures)

4.4 Speed Sensitivity Procedures

A subset of test conditions from Free-Rolling and Brake/Drive tests were repeated in separate tests, in which the roadway speed was varied.

4.5 Longitudinal Transient Procedures

Although sweeps for longitudinal transients were included in the Brake/Drive Force & Moment sections, additional tests were done to provide more detailed data. Two test types were run:

1. RelaxSRSine – Sine sweeps of slip ratio at different conditions
2. Static Longitudinal Stiffness – provides data of static tire carcass stiffness through precise measurement of belt (roadway) travel distance

4.6 Track Replay Procedures

A section ("The Grate") of the Mt. Panorama track was simulated on the tire testing machine. This was done by continuously varying all tire inputs in accordance with what is observed on the race vehicle. The section was repeated multiple times with slightly varied input conditions. Note that for this test procedure, the warm-up was run as a separate test run immediately prior to the track replay run.

5. Test Procedure Details

There are a vast number of methods to measure tire performance at Calspan. However, given the popularity and well-known structure, the tests have been designed to enable the creation of a Pacejka type tire model.

The creation of a full Pacejka model requires both Free-Rolling and Brake/Drive tests to be performed. The Free-Rolling test determines the pure lateral capability of the tire by steering it (sweeping slip angle) at speed and measuring the resulting forces and moments. This is performed at different combinations of vertical load, inclination angle, slip angle, and inflation pressures. The Brake/Drive test does the same as the above using the same test conditions. However, this time the tire is held at a fixed steer angle (slip angle) and slip ratio is being swept allowing for pure and combined forces and moments to be measured.

Roadway speed also creates another variable however, for this testing most measurements were done at a test speed of 100 or 80 km/h, depending on test type. This speed is close to an average apex speed for the race vehicle. The effects of roadway speed on tire performance were explored in separate tests.

The commanded test conditions can also be investigated by looking at the commanded channels in the data files. The commanded channels can be identified through the prefix "C", for example "CSA".

5.1 Test Conditions

The test conditions for the procedures were selected based on current vehicle data and setup information. The table below shows which conditions that have been used in testing, with the parameter being swept highlighted in orange. Note that not all combinations of all parameters have been evaluated.

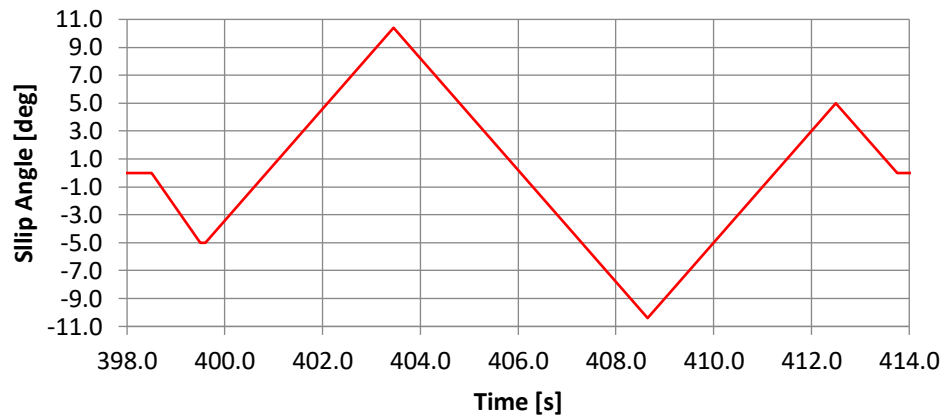
Free Rolling Variables						
Road Speed	Slip Angle	Load	Camber	Pressure	Slip Ratio	
V (m/s)	SA (Degrees)	Fz (N)	IA (Degrees)	P (psi)	SR (-)	
27.77	10 (max)	8500	-9.0	21	0	
	-10 (min)	6250	-7.5	24		
	Rate	4000	-6.0	27		
	4 degrees/s	1750	-4.5	30		
		500	-3.0	33		
			-1.5			
			0.0			
Combined / Brake and Drive Variables						
Road Speed	Slip Angle	Load	Camber	Pressure	Slip Ratio	Notes
V (m/s)	SA (Degrees)	Fz (N)	IA (Degrees)	P (psi)	SR (-)	
22.22	-4	8500	-9.0	21	-0.30 (min)	+/-30%
	-2.5	6250	-7.5	24	0.30 (max)	
	0	4000	-6.0	27	Rate	
	2.5	1750	-4.5	30	Variable	
	5	500	-3.0	33		
			-1.5			
			0.0			

5.2 Warm-Up

In the warm-up the tire is put through a series of working conditions until a stable and consistent force and moment output is observed. Two check tests on the warm-up procedures were performed to ensure the tire was sufficiently broken-in and warmed-up. This warm-up is then implemented across all the test procedures at the middle pressure (27 psi).

5.3 SC_FreeRoll

This is the main Force & Moment section for the free-rolling tests. At the beginning of the test, procedures to determine lateral relaxation length are done using step-steer and sine-wave methods. The unloaded radius is also measured. After that, a large number of combinations of load and camber are evaluated using slip angle sweeps. The profile of the slip angle sweep is shown below.



The slip angle sweep is repeated for a number of different test conditions, shown in the matrix below. Note that not all combinations are tested, as some are unlikely to occur on-track.

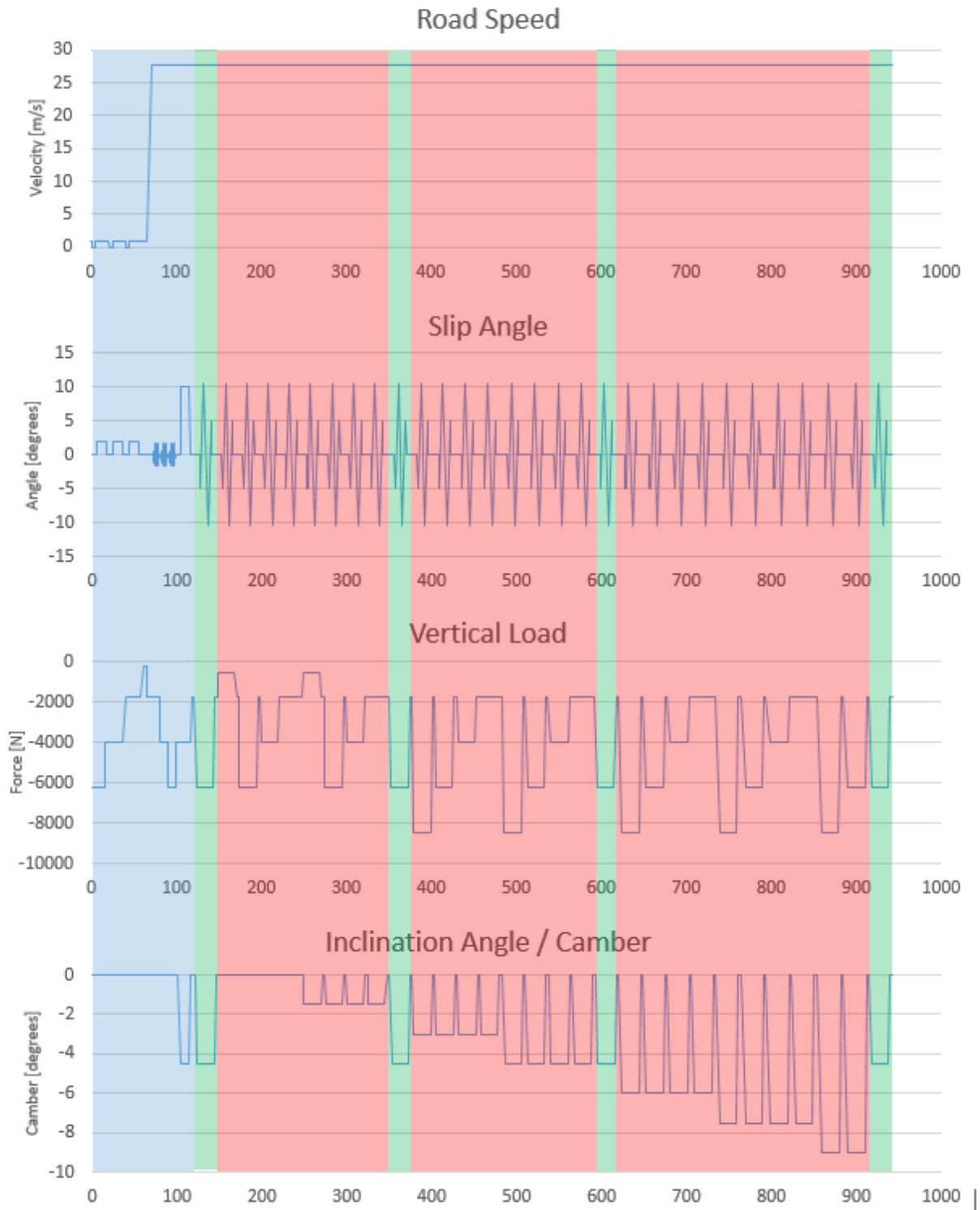
Matrix							
Fz [N] / IA [deg]	-9.0	-7.5	-6.0	-4.5	-3.0	-1.5	0.0
8500	1	1	1	1	1	0	0
6250	1	1	1	1	1	1	1
4000	0	1	1	1	1	1	1
1750	0	1	1	1	1	1	1
500	0	0	0	0	0	1	1

Below are charts of the entire test procedure, color-coded to represent the sweep types taking place. Time is shown on the x-axis in seconds. In-between slip angle sweeps the tire always returns to the thermal pause conditions to help ensure that the tire starts the next sweep at the target temperature.

Blue: Step Steer, Sine Wave, Unloaded Radius, and Steer Hold Test

Green: Check Sweep Test

Red: Matrix Test



5.4 Vertical Stiffness

The vertical stiffness test is performed directly following the Free-Rolling test segment. This test puts the tire through a range of different conditions over a short period of time. This is to provide measurements of the loaded and effective rolling radius of the tire at as many conditions as possible. This data allows for calculation of the vertical stiffness sensitivity with respect to parameters such as speed, lateral force, load, pressure and camber.

5.5 Thermal

The thermal test is performed directly following the Free-Rolling test segment. Slip angle, load, camber, and pressure are kept constant while the roadway speed is repeatedly ramped up and down. From this, the effects of tread temperature and sliding velocity on tire friction can be evaluated.

5.6 IA Sweep and ExtraIA

The IA Sweep (camber sweep) is performed directly following the Free-Rolling test segment. This test stepwise sweeps the camber while road speed, slip angle, load, and pressure stay the same. The sweep is repeated multiple times at two different loads and several pressures, which gives a more detailed view of camber sensitivity than the main force and moment test. The “ExtraIA” test follows a similar structure, but is done at higher slip angles and for a larger range of inclination angles.

5.7 FreeRoll with Different Speeds

SC_FreeRoll_wDifferentSpeeds is essentially a reduced version of the free rolling test in terms of test conditions with only one pressure and three cambers. However, the speed of the test is changed to evaluate the effect of speed on tire performance. The test conditions are shown below.

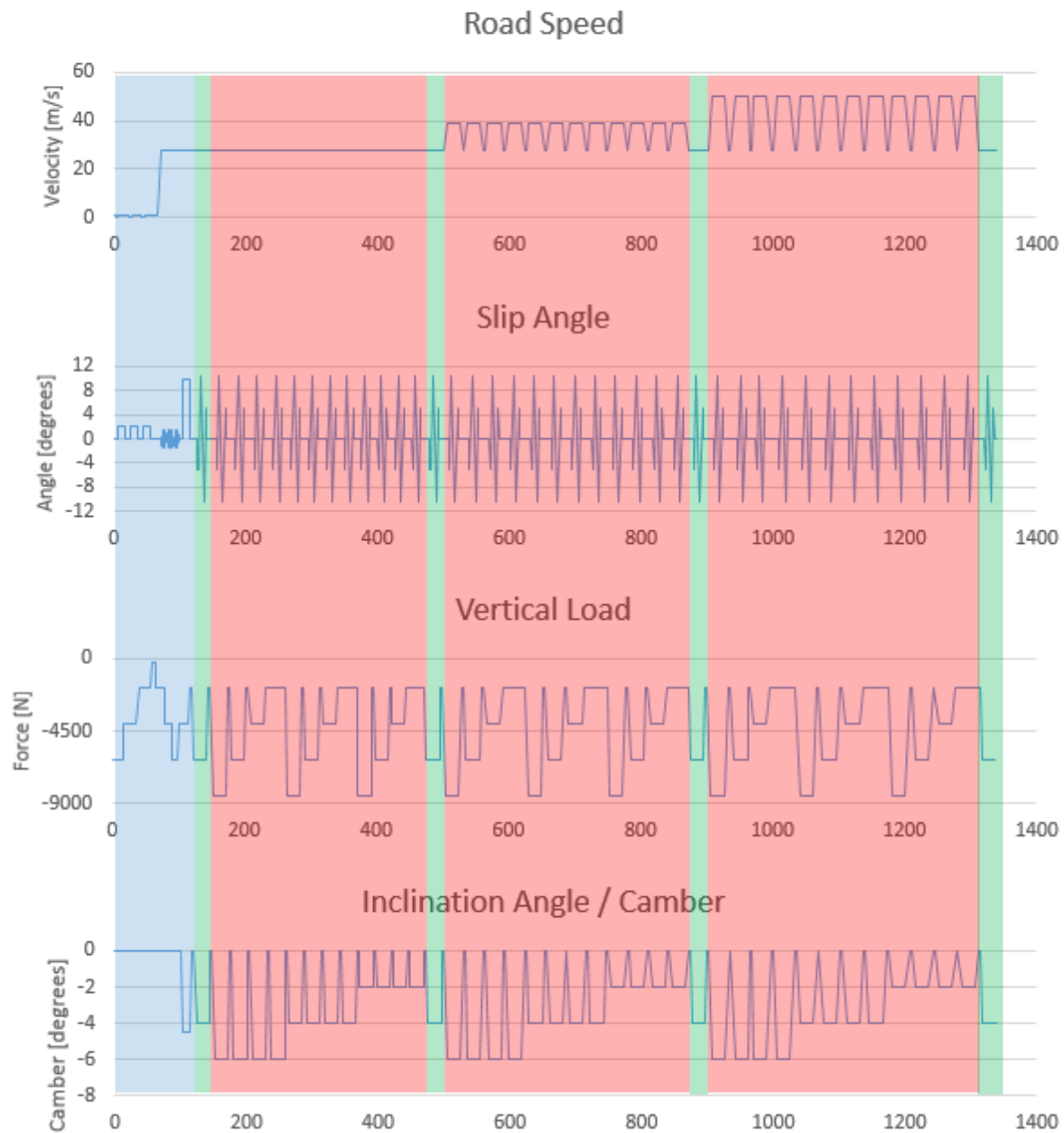
SC_FreeRolling_wDifferentSpeeds					
Road Speed	Slip Angle	Load	Camber	Pressure	Slip Ratio
V (m/s)	SA (Deg)	Fz (N)	IA (Deg)	P (psi)	SR (-)
27.77	10 (max)	8500	-6.0	30	0
38.89	-10 (min)	6250	-4.5		
50.00	Rate	4000	-3.0		
	4 deg/s	1750			

Below are charts of the entire test procedure, color-coded to represent the sweep types taking place. Time is shown on the x-axis in seconds. In-between slip angle sweeps the tire always returns to the thermal pause conditions to help ensure that the tire always starts the next sweep at the same temperature helping to keep results consistent.

Blue: Step Steer, Sine Wave, Unloaded Radius, and Steer Hold Test

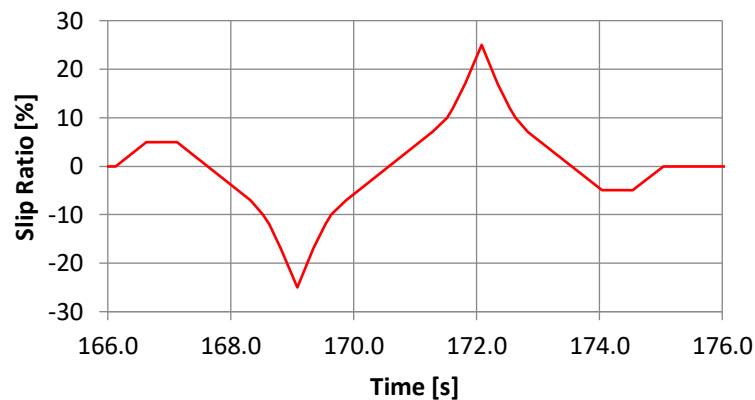
Green: Check Sweep Test

Red: Matrix Test



5.8 SC_BrakeDrive

This is the main Force & Moment section for the Brake/Drive tests. There are no additional tests at the beginning or end of this procedure aside from warm-up. In this test a large number of load, camber and slip angle conditions are evaluated using slip ratio sweeps. The profile of the slip ratio sweep can be seen below.



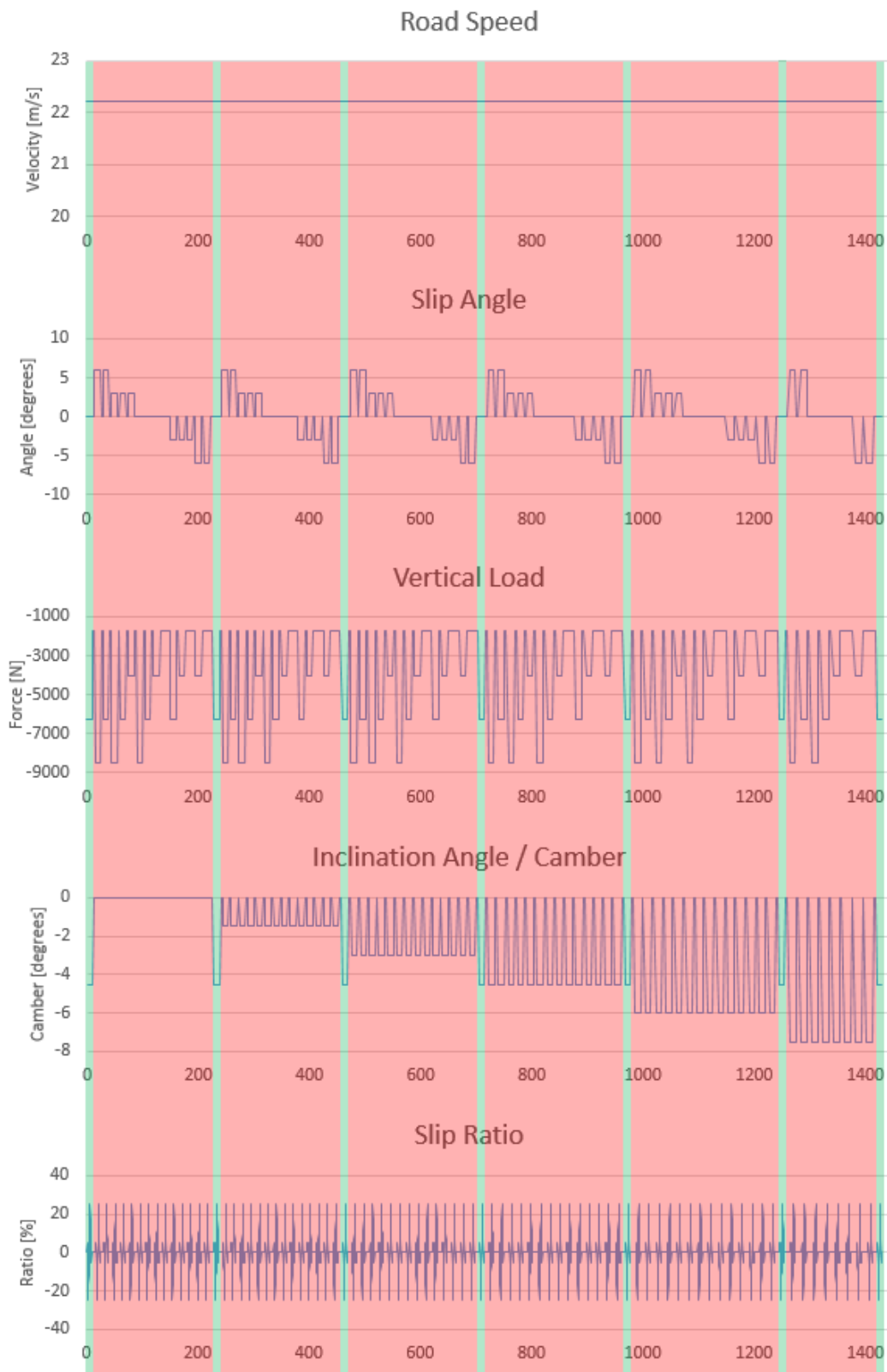
The slip ratio sweep is repeated for a number of different test conditions, shown in the matrix below. Note that not all combinations are tested as some are unlikely to occur on-track. The highest camber condition (-7.5 deg) was tested at an even further reduced matrix. The slip angle values selected for the Brake/Drive tests were based on the results of the free-rolling tests.

Fz [N]/ SA [°]	Matrix (IA > -6°)					Matrix (IA = -7.5°)				
	5	2.5	0	-2.5	-4	5	2.5	0	-2.5	-4
8500	0	0	1	1	1	0	0	1	0	1
6250	0	1	1	1	1	0	0	1	0	1
4000	1	1	1	1	0	1	0	1	0	0
1750	1	1	1	0	0	1	0	1	0	0

Below are charts of the entire test procedure, color-coded to represent the sweep types taking place. Time is shown on the x-axis in seconds. In-between slip ratio sweeps the tire always returns to the thermal pause conditions to help ensure that the tire always starts the next sweep at the same temperature.

Green: Check Sweep Test

Red: Matrix Test



5.9 Brake/Drive with Different Speeds

SC_BrakeDrive_wDifferentSpeeds is essentially a reduced version of the Brake/Drive test in terms of test conditions with only one pressure and three cambers. However, the speed of the test is changed to evaluate the effect of speed on tire performance. The test conditions are shown below.

SC_BrakeDrive_wDifferentSpeeds					
Road Speed	Slip Angle	Load	Camber	Pressure	Slip Ratio
V (m/s)	SA (Deg)	Fz (N)	IA (Deg)	P (psi)	SR (-)
22.22	-4.0	8500	-6.0	30 psi	-0.3 (-30%)
33.33	-2.5	6250	-4.5		0.3 (30%)
44.44	0.0	4000	-3.0		
	2.5	1750			Rate
	5.0				Variable

For each speed and camber the following test conditions have been tested:

Matrix					
Fz [N] / SA [°]	-4.0	-2.5	0.0	2.5	5.0
8500	0	0	1	1	1
6250	0	1	1	1	1
4000	1	1	1	1	1
1750	1	1	1	1	0
500	1	1	1	0	0
